

Project Initialization and Planning Phase

Date	27 June 2024
Team ID	xxxxxx
Project Title	OptiCrop – AI-Based Crop Recommendation System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed OptiCrop – AI-Based Crop Recommendation System aims to help farmers choose the most suitable crop based on soil nutrients and weather conditions using Machine Learning. The system predicts the best crop by analyzing parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall.

Project Overview	
Objective	The primary objective is to develop a Machine Learning-based crop recommendation system that provides accurate crop suggestions through a simple web application.
Scope	The project focuses on collecting soil and environmental parameters, preprocessing the data, applying a trained Machine Learning model, and displaying the recommended crop using a Flask-based web interface.
Problem Statement	
Description	Farmers often face difficulty in selecting the right crop due to limited knowledge of soil nutrients and weather conditions, leading to poor crop productivity.

Impact	Incorrect crop selection can reduce agricultural productivity, increase financial losses, and negatively affect farmers' income.
Proposed Solution	
Approach	Develop a Machine Learning model using Scikit-learn, integrate it with a Flask backend, preprocess soil and weather data, and recommend the most suitable crop through a responsive web application.
Key Features	AI-based crop recommendation system,Real-time crop prediction using Flask,User-friendly interface for entering soil and weather parameters.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU specifications	Intel Core i5 or above
Memory	RAM specifications	8 GB
Storage	Space for dataset, models and project files.	10 GB Free Disk Space
Software		
Frameworks	Backend Framework	Flask
Libraries	Machine Learning Libraries.	scikit-learn, pandas, Numpy, Joblib
Development Environment	IDE	Visual Studio Code
Data		
Data	Crop Recommendation Dataset	Kaggle Crop Recommendation Dataset (CSV Format, 2200 Records)